Mid-Semester Examination – Evaluation Scheme

Question 6

A 0.5 m length of a coaxial cable has inner conductor radius $a = 1.0$ mm and outer conductor radius $b = 4.0$ mm. The inner conductor carries a total charge of $Q = 30$ nC. Find:

(a) The line charge density (ρ_L).

(b) The surface charge density (ρ_s) on the inner and outer conductors.

(c) The electric field $\mathbf{E}$ and flux density $\mathbf{D}$ for $a < \rho < b$.

(d) The electric field $\mathbf{E}$ and flux density $\mathbf{D}$ for $\rho < a$ and $\rho > b$.

Solution

Given: $L = 0.50$ m, $a = 1.0 \times 10^{-3}$ m, $b = 4.0 \times 10^{-3}$ m, $Q = 30 \times 10^{-9}$ C, $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m.

(a) Line charge density

$$\rho_L = \frac{Q}{L} = \frac{30 \times 10^{-9}}{0.50} = 60.0 \text{ nC/m}.$$

(b) Surface charge density on cylindrical surfaces

$$\rho_s(\rho) = \frac{\rho_L}{2\pi\rho}.$$

At inner radius a:

$$\rho_s(a) = \frac{\rho_L}{2\pi a} = \frac{6.0 \times 10^{-8}}{2\pi(1.0 \times 10^{-3})} \approx 9.55 \ \mu\text{C/m}^2.$$

At outer radius b:

$$\rho_s(b) = \frac{-\rho_L}{2\pi b} = \frac{-6.0 \times 10^{-8}}{2\pi(4.0 \times 10^{-3})} \approx -2.39 \ \mu\text{C/m}^2.$$

(c) Electric field and flux density for $a < \rho < b$

The internal fields may therefore be calculated easily:

$$D_\rho = \frac{a\,\rho_s}{\rho} = \frac{10^{-3}(9.55 \times 10^{-6})}{\rho} = \frac{9.55}{\rho}\hat{a}_\rho \text{ nC/m}^2$$

and

$$E_\rho = \frac{D_\rho}{\varepsilon_0} = \frac{9.55 \times 10^{-9}}{8.854 \times 10^{-12}\,\rho} = \frac{1079}{\rho}\hat{a}_\rho \text{V/m}.$$

(d) Regions $\rho < a$ and $\rho > b$

$$\mathbf{E} = \mathbf{D} = \mathbf{0} \quad \text{for } \rho < a \text{ and } \rho > b.$$

Marking Scheme

(a) Line charge density ρ_L: **1 mark**.

(b) Surface charge densities $\rho_s(a)$ and $\rho_s(b)$: **1 mark**.

(c) Expressions for $\mathbf{E}$ and $\mathbf{D}$ in $a < \rho < b$: **1 mark**.

(d) Fields in $\rho < a$ and $\rho > b$: **1 mark**.

Partial marking is awarded depending on correctness and steps.

Question 7

Three infinite uniform sheets of charge are located in free space as follows:
3 nC/m^2 at $z = -4$ m, 6 nC/m^2 at $z = 1$ m, and -8 nC/m^2 at $z = 4$ m.
Find the electric field vector $\mathbf{E}$ at the point:
(a) $P(-1, -5, 2)$; (b) $Q(-2, 4, -2)$.

Solution

Step 1: Formula.
For an infinite sheet of charge density ρ_s at $z = z_0$, the electric field contribution at a point with coordinate z is

$$E_{z,\text{sheet}} = \frac{\rho_s}{2\varepsilon_0}\,\text{sign}(z - z_0),$$

where sign$(x) = +1$ if $x > 0$ and -1 if $x < 0$. Direction is along $\hat{a}_z$.

Step 2: Convert charge densities.

$$\begin{aligned} \rho_{s1} &= +3 \times 10^{-9}\,\text{C/m}^2, & z_1 &= -4 \\ \rho_{s2} &= +6 \times 10^{-9}\,\text{C/m}^2, & z_2 &= 1 \\ \rho_{s3} &= -8 \times 10^{-9}\,\text{C/m}^2, & z_3 &= 4 \end{aligned}$$

Magnitudes:

$$\frac{\rho_{s1}}{2\varepsilon_0} = 169.4\,\text{V/m}, \quad \frac{\rho_{s2}}{2\varepsilon_0} = 338.8\,\text{V/m}, \quad \frac{\rho_{s3}}{2\varepsilon_0} = -451.8\,\text{V/m}.$$

(a) Point $P(-1,-5,2)$, i.e. $z=2$.

$$\begin{aligned}
\text{For } z_1=-4: &\quad \text{sign}(2-(-4))=+1 &&\Rightarrow +169.4\\
\text{For } z_2=1: &\quad \text{sign}(2-1)=+1 &&\Rightarrow +338.8\\
\text{For } z_3=4: &\quad \text{sign}(2-4)=-1 &&\Rightarrow (-451.8)(-1)=+451.8
\end{aligned}$$

$$E_z(P)=169.4+338.8+451.8=960.0 \text{ V/m}$$

Thus,

$$\mathbf{E}_P=960\ \hat{a}_z \text{ V/m}.$$

(b) Point $Q(-2,4,-2)$, i.e. $z=-2$.

$$\begin{aligned}
\text{For } z_1=-4: &\quad \text{sign}(-2-(-4))=+1 &&\Rightarrow +169.4\\
\text{For } z_2=1: &\quad \text{sign}(-2-1)=-1 &&\Rightarrow (338.8)(-1)=-338.8\\
\text{For } z_3=4: &\quad \text{sign}(-2-4)=-1 &&\Rightarrow (-451.8)(-1)=+451.8
\end{aligned}$$

$$E_z(Q)=169.4-338.8+451.8=282.4 \text{ V/m}$$

Thus,

$$\mathbf{E}_Q=282.4\ \hat{a}_z \text{ V/m}.$$

Final Answers:

$$\begin{aligned}
(a)\quad &\mathbf{E}_P=960\ \hat{a}_z \text{ V/m}\\
(b)\quad &\mathbf{E}_Q=282.4\ \hat{a}_z \text{ V/m}
\end{aligned}$$

Marking Scheme:

- Part (a): 1.5 marks (full correct answer). Partial marks can be given based on correctness of method and intermediate steps.
- Part (b): 1.5 marks (full correct answer). Partial marks can be given based on correctness of method and intermediate steps.

Question 8

Calculate the total electric flux leaving the cubical surface formed by the six planes $x, y, z = \pm 5$ m if the charge distribution is:

(a) Two point charges, 0.1 μC at $(1,-2,3)$ and $\frac{1}{7}$ μC at $(-1,2,-2)$.

(b) A uniform line charge of π μC/m at $x=-2$ m, $y=3$ m.

Solution

Gauss's law in terms of electric flux density D:

$$\Phi_D = \iint_S \mathbf{D} \cdot d\mathbf{A} = Q_{\text{enc}}^{(\text{free})}.$$

When using $\mathbf{D}$, the flux equals the total enclosed free charge (no division by ε_0).

(a) Two Point Charges

Both charges $(1, -2, 3)$ and $(-1, 2, -2)$ lie inside the cube $(x, y, z = \pm 5)$.

$$\begin{aligned} Q_{\text{enc}} &= 0.1\ \mu\text{C} + \tfrac{1}{7}\ \mu\text{C} \\ &= \left(0.1 + \tfrac{1}{7}\right)\mu\text{C} \\ &= \tfrac{17}{70}\ \mu\text{C} \\ &= \tfrac{17}{70} \times 10^{-6}\ \text{C} \\ &= 0.243\ \mu\text{C} \\ &= 0.243 \times 10^{-6}\ \text{C (approx)}. \end{aligned}$$

Therefore,

$$\Phi_D = Q_{\text{enc}} = 0.243 \times 10^{-6}\ \text{C}.$$

(b) Uniform Line Charge

Linear charge density:

$$\rho_L = \pi\ \mu\text{C/m} = \pi \times 10^{-6}\ \text{C/m}.$$

The line extends parallel to the z-axis at $x = -2$, $y = 3$. Inside the cube, z ranges from -5 to $+5$, so $L = 10$ m.

$$\begin{aligned} Q_{\text{enc}} &= \rho_L L = (\pi \times 10^{-6})(10) \\ &= \pi \times 10^{-5}\ \text{C} \\ &\approx 31.42 \times 10^{-6}\ \text{C}. \end{aligned}$$

Therefore,

$$\Phi_D = Q_{\text{enc}} = 31.42 \times 10^{-6}\ \text{C}.$$

Final Answers:

$$\begin{aligned} &(a) \quad \Phi_D = 0.243 \times 10^{-6}\ \text{C} \\ &(b) \quad \Phi_D = 31.42 \times 10^{-6}\ \text{C} \end{aligned}$$

Marking Scheme:

- Part (a): 1.5 marks (full correct answer). Partial marks can be given based on correctness of method and intermediate steps.
- Part (b): 1.5 marks (full correct answer). Partial marks can be given based on correctness of method and intermediate steps.

9) Statement

$$\oint_S D \cdot ds = \int_V (\nabla \cdot D)\, dV \longrightarrow (1M)$$

+ Figure

Proof: $\oint D \cdot ds = Q = \int_{front} + \int_{back} + \int_{Left} + \int_{right} + \int_{top} + \int_{bottom}$

$\hookrightarrow (1M)$

$$\int_{front} + \int_{back} = \frac{\partial D_x}{\partial x} \Delta x \Delta y \Delta z$$

$$\left[\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right] = \frac{\oint D \cdot ds}{\Delta v} = \rho_v \quad -(1M)$$

$$\int_V (\nabla \cdot D)\, dv = \oint_S D \cdot ds$$

10) $\nabla \cdot D = \frac{1}{\rho} \frac{\partial}{\partial \rho}(\rho^2 z^2 \sin^2\phi + \frac{1}{\rho}\frac{\partial}{\partial \phi}(\rho z^2 \sin\phi\cos\phi)$

$+ \frac{\partial}{\partial z}(\rho^2 z \sin^2\phi) \longrightarrow (1M)$

$= 2z^2 \sin^2\phi + z^2\cos 2\phi + \rho^2 \sin^2\phi$

$\nabla \cdot D = z^2 + \rho^2 \sin^2\phi \ C/m^3 \quad -(1M)$

Q.11. $V = 80\phi$

$$\vec{E} = -\nabla V = -\frac{1}{\rho}\frac{\partial V}{\partial \phi}\hat{a}_\phi = -\frac{80}{\rho}\hat{a}_\phi \; V/m.$$

Energy density, $w_E = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 \cdot \frac{80^2}{\rho^2}$. — (0.5M)

a) $W_E = \int_{vol.} w_E \, dv = \int_{vol.} w_E \, \rho \, d\rho \, d\phi \, dz$

$$= \int_0^1 \int_0^{0.2\pi} \int_{2cm}^{4cm} \frac{1}{2} \cdot \varepsilon_0 \cdot \frac{80^2}{\rho^2} \, \rho \, d\rho \, d\phi \, dz$$

— (1M)

$$= 12.3n\,J$$

b) $V_{AB} = -\int_B^A \vec{E} \cdot d\vec{l} = -\int_{0.2\pi}^{0} -\frac{80}{\rho}\hat{a}_\phi \cdot \rho \, d\phi \, \hat{a}_\phi$

— (1M)

$$= -16\pi\,V = -50.2\,V$$

c) $w_E \propto \frac{1}{\rho^2} \Rightarrow w_{E(max)}$ at $(\rho_{min.} = 2cm)$

$$w_{E(max)} = \frac{1}{2}\varepsilon_0 \frac{80^2}{(2cm)^2} = 70.8\,\mu J/m^3$$ — (0.5M)

Q.12). $\vec{E} = (3x^2 + y)\bar{a}_x + x\,\bar{a}_y$ kV/m , $Q = -2\mu C$.

$$W = -Q \int_{(0,5,0)}^{(2,-1,0)} \vec{E} \cdot d\vec{l}$$

$$= -Q \int_0^2 (3x^2 + y)\,dx - Q \int_5^{-1} x\,dy \quad \text{—— (1 M)}$$

$$= -Q \times 6 \times 10^3 = \underline{\underline{12\ mJ}} \quad \text{—— (1 M)}.$$

(13) $D_{N_2} = 70 a_2 \ nc/m^2$

$D_{t_2} = -18.75 a_n + 31.25 a_y \ nc/m^2$

Because, $D_{N_1} = 70 a_n \ nc/m^2$ and D_{t_1} is $-30 a_n + 50 a_y \ nc/m^2$

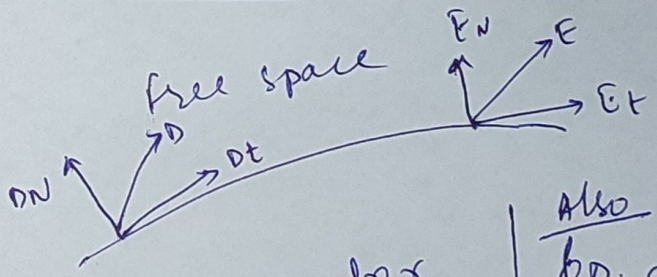
As boundary condition of Dielectric - Dielectric media is

for $\rho_s = 0$; $\boxed{D_{N_1} = D_{N_2}}$

&

$$\boxed{\frac{D \tan 1}{D \tan 2} = \frac{\varepsilon_1}{\varepsilon_2}}$$

(14)



$\oint E \cdot dl = 0$

$\therefore E_t \Delta w - E_N \frac{1}{2} \Delta h + E_N \frac{1}{2} \Delta h = 0$

$\boxed{E_t = 0}$

Also

$\oint D \cdot ds = Q$

$D_N \Delta s = Q = \rho_s \Delta s$

$\boxed{D_N = \rho_s}$

$D_t = E_t = 0$; $D_N = \varepsilon_0 E_N = \rho_s$